

Supplementary Materials: Evaluating epidemiological forecasts using daily versus weekly aggregated data

This document provides provincial-level results for all provinces and the national aggregate not shown in the main text. The Eastern Cape (EC) is presented as the representative example in the main paper; results for all other provinces are shown below.

Forecast performance

Free State (FS)

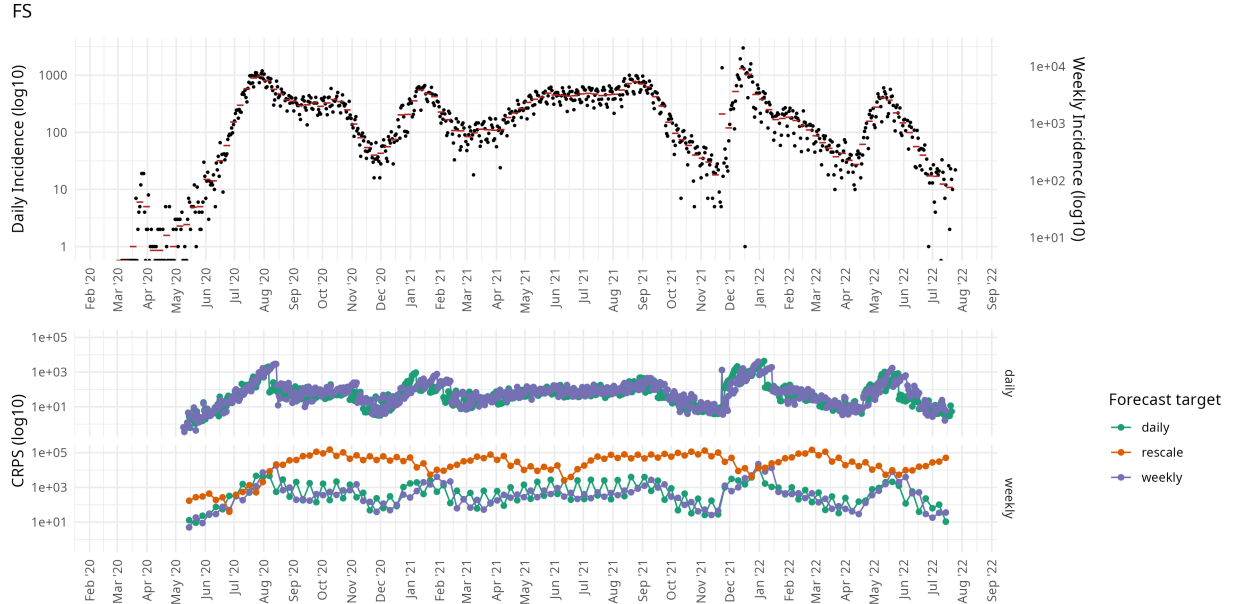


Figure 1: Forecast performance for the Free State province (FS). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

Gauteng (GP)

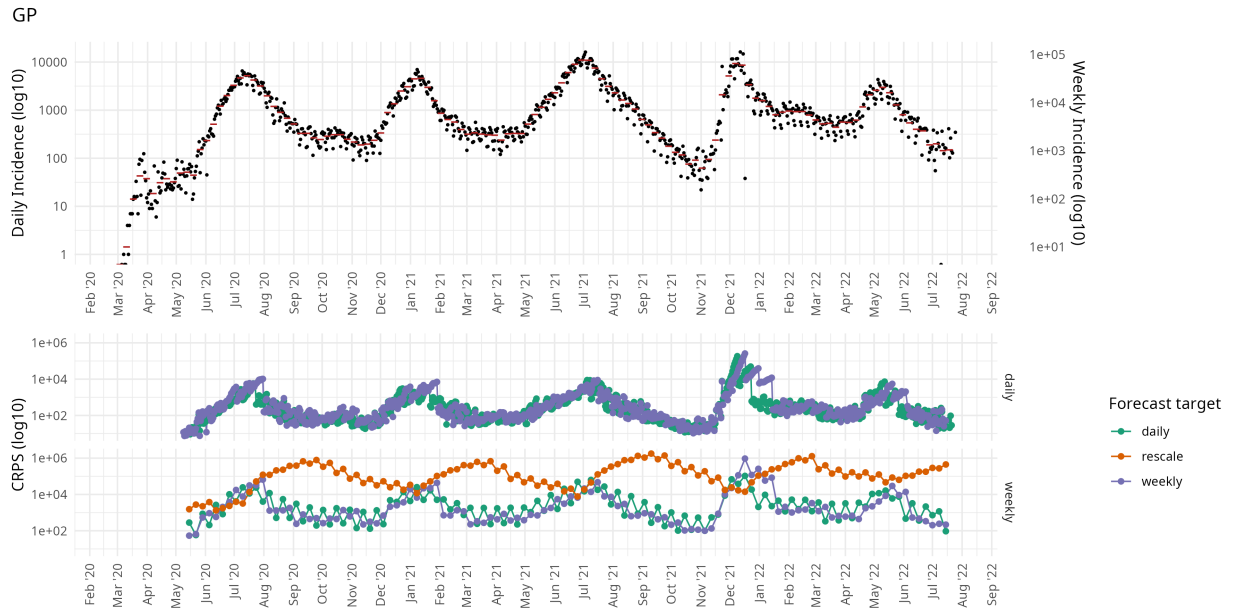


Figure 2: Forecast performance for the Gauteng province (GP). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

KwaZulu-Natal (KZN)

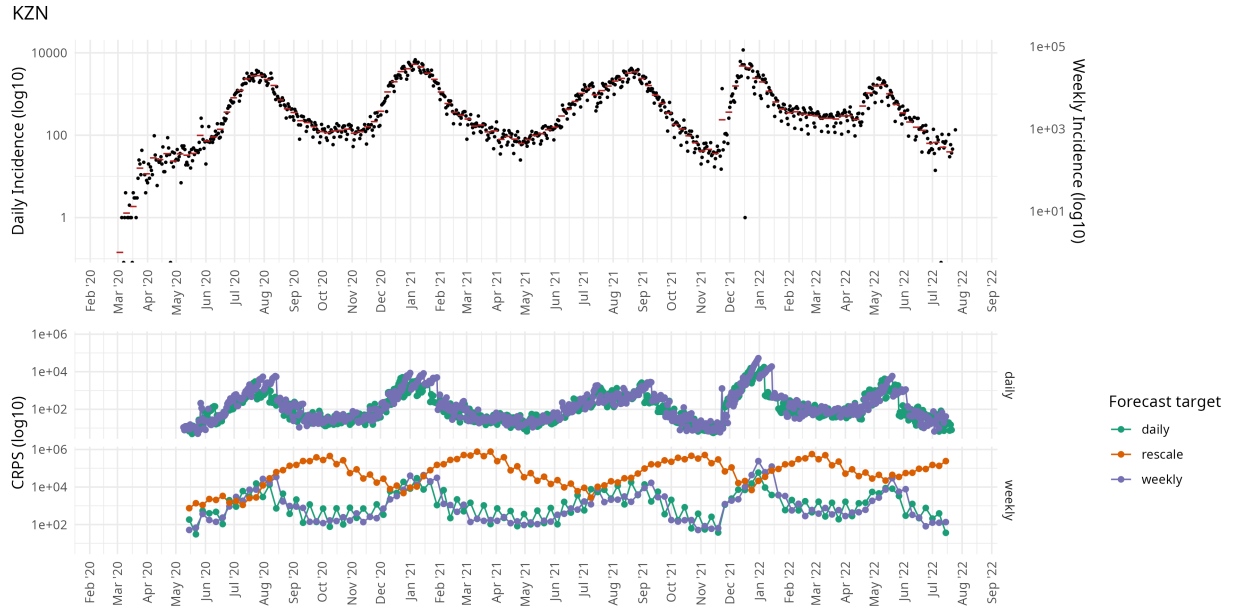


Figure 3: Forecast performance for the KwaZulu-Natal province (KZN). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

Limpopo (LP)

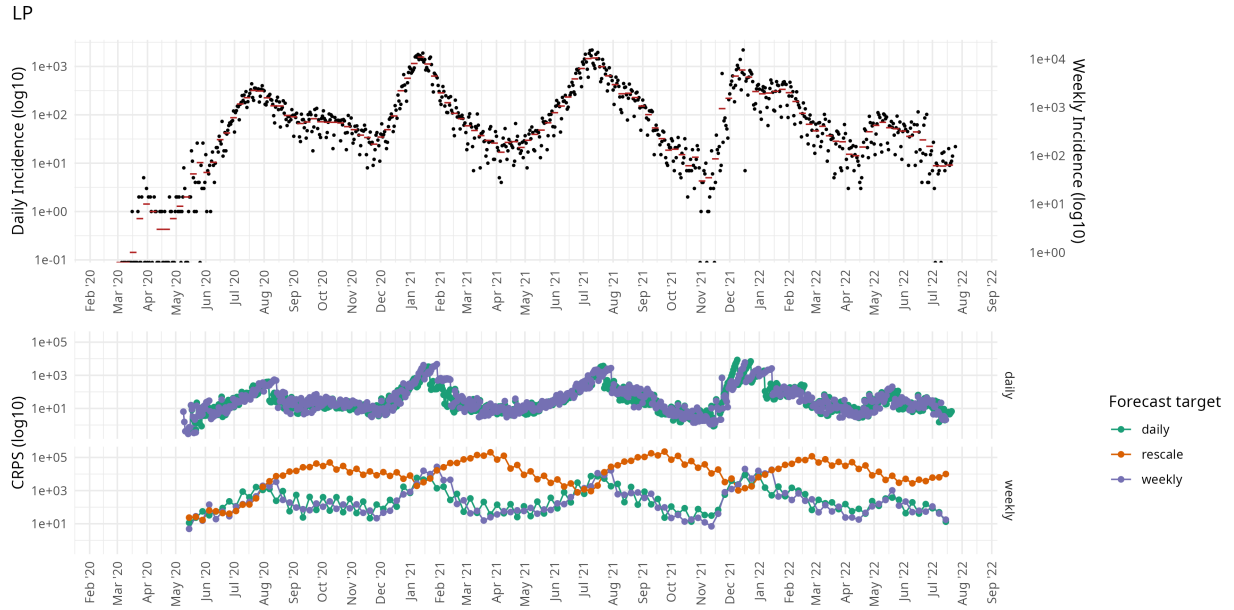


Figure 4: Forecast performance for the Limpopo province (LP). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

Mpumalanga (MP)

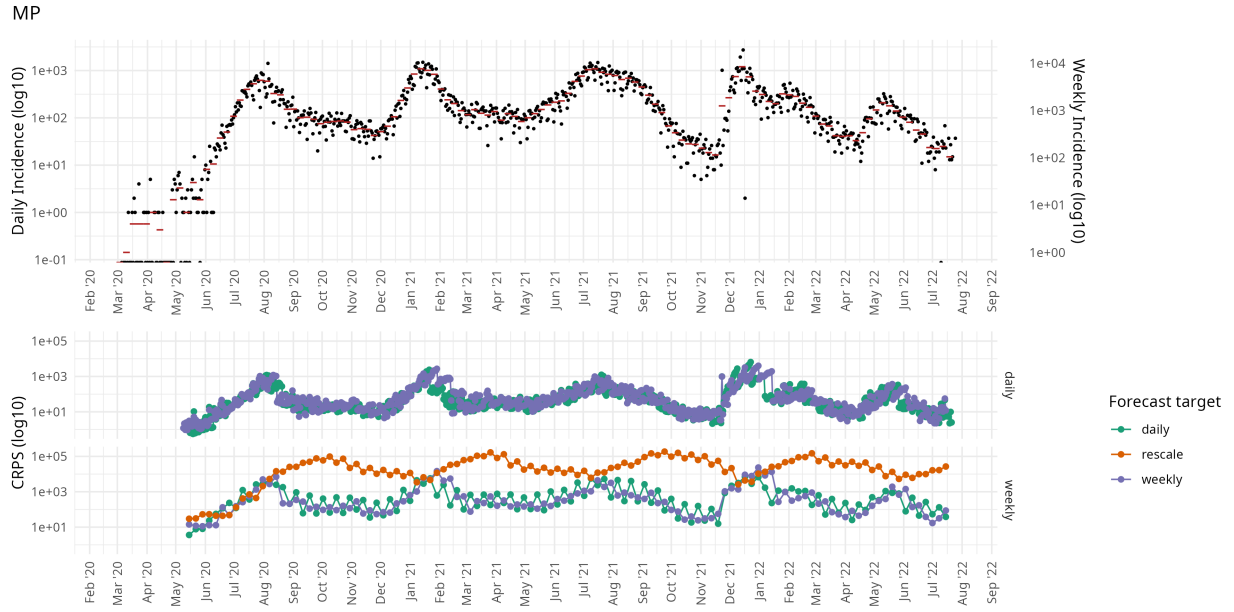


Figure 5: Forecast performance for the Mpumalanga province (MP). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

Northern Cape (NC)

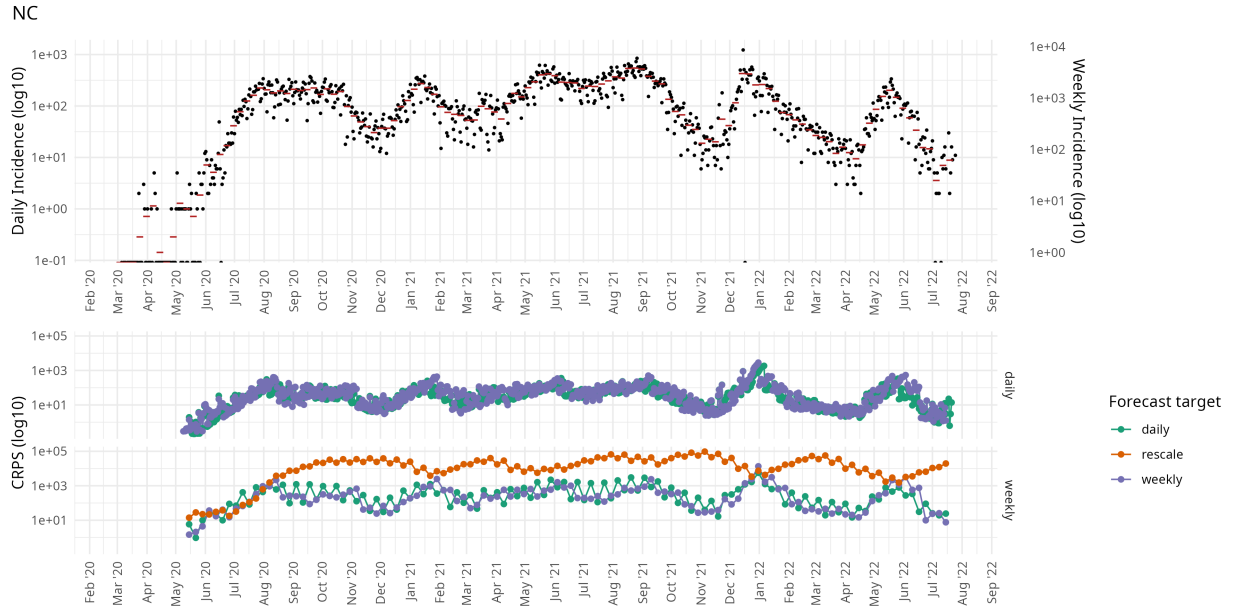


Figure 6: Forecast performance for the Northern Cape province (NC). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

North West (NW)

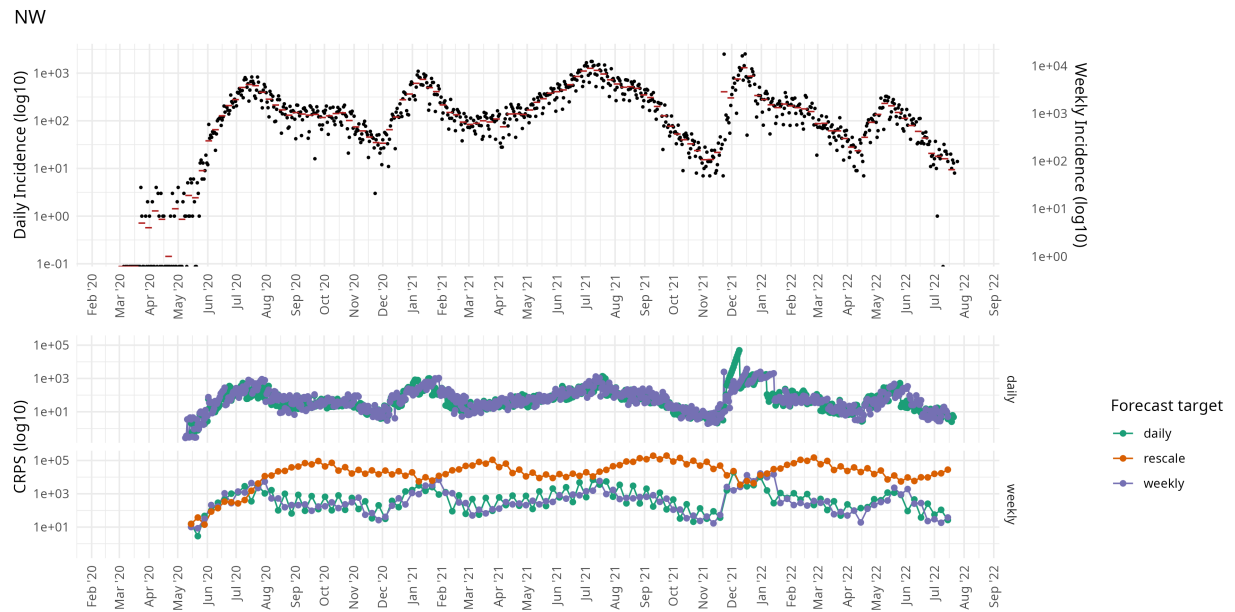


Figure 7: Forecast performance for the North West province (NW). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

South Africa national aggregate (RSA)

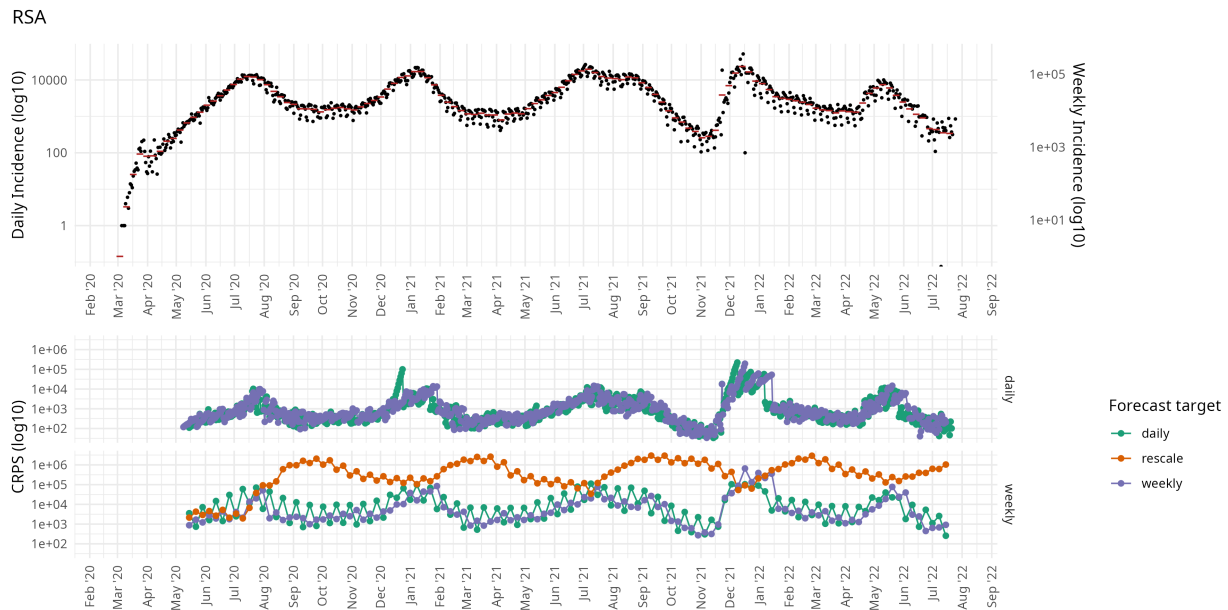


Figure 8: Forecast performance for the national aggregate (RSA). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

Western Cape (WC)

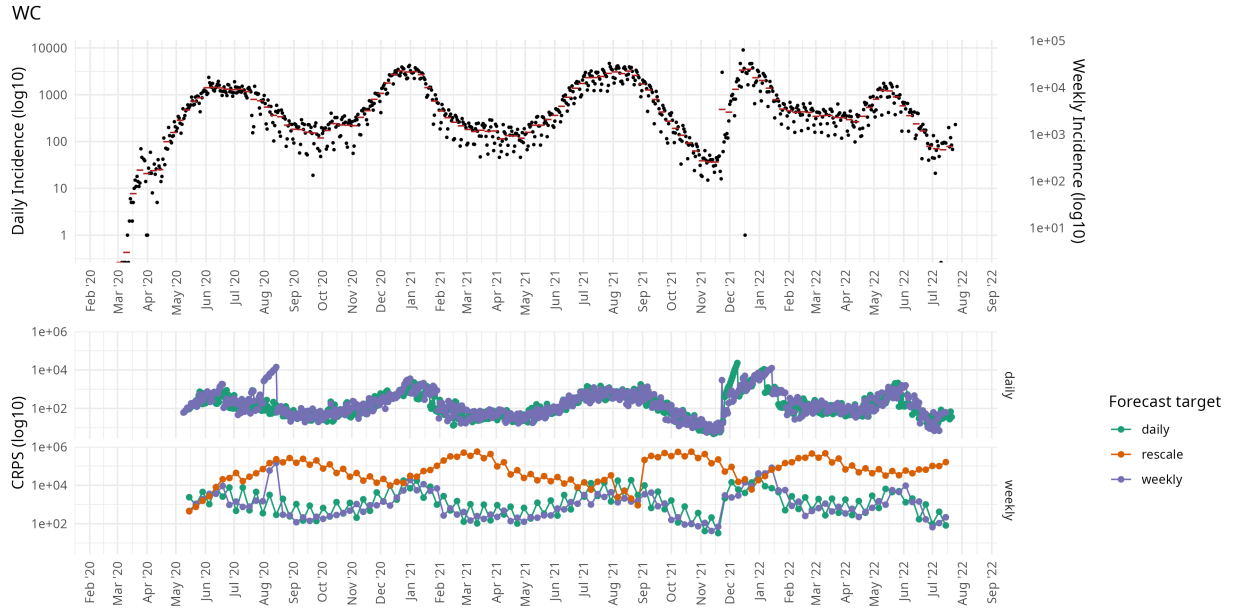


Figure 9: Forecast performance for the Western Cape province (WC). The top panel shows the daily (black points) and weekly (red horizontal segments) COVID-19 incidence. The bottom panel displays the Continuous Ranked Probability Score (CRPS) for daily, weekly, and rescaled weekly forecast targets (colours), with both incidence and CRPS shown on a log10 scale. The rows in the bottom panel are faceted by the temporal resolution of the evaluation data (daily or weekly).

Computational diagnostics

Free State (FS)

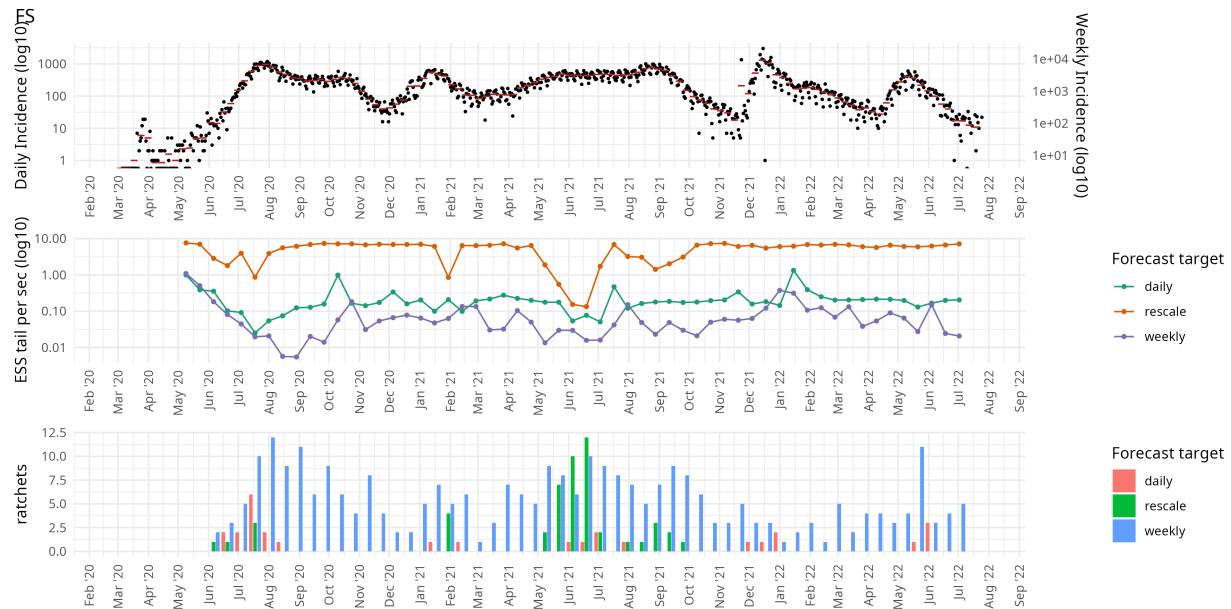


Figure 10: Computational diagnostics for the Free State province (FS). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

Gauteng (GP)

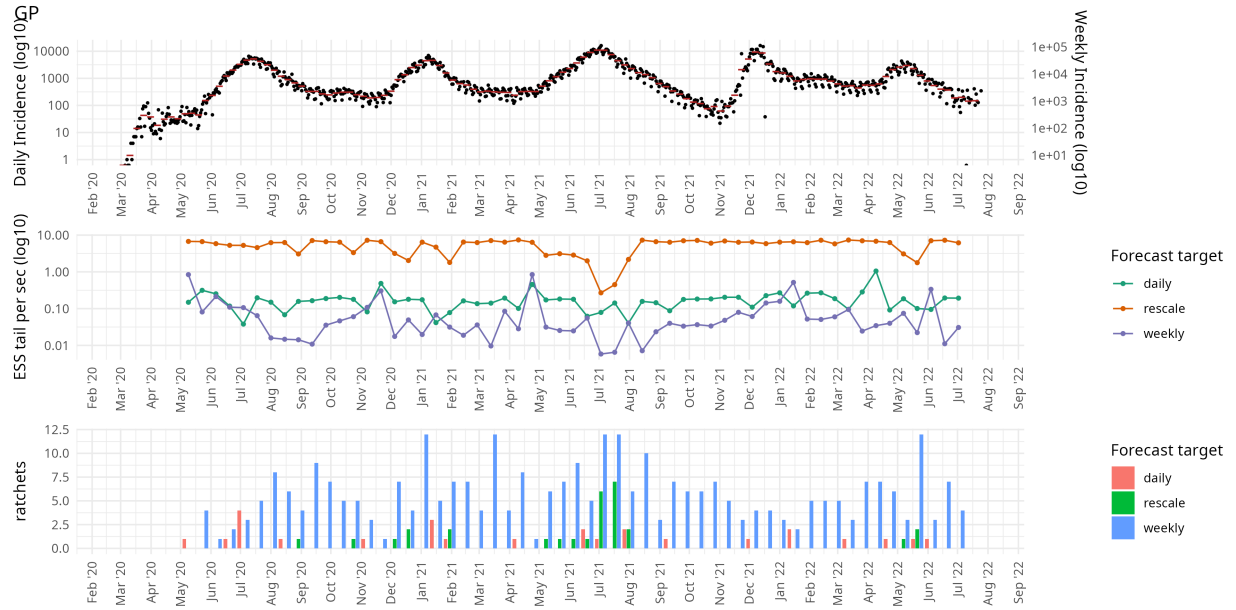


Figure 11: Computational diagnostics for the Gauteng province (GP). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

KwaZulu-Natal (KZN)

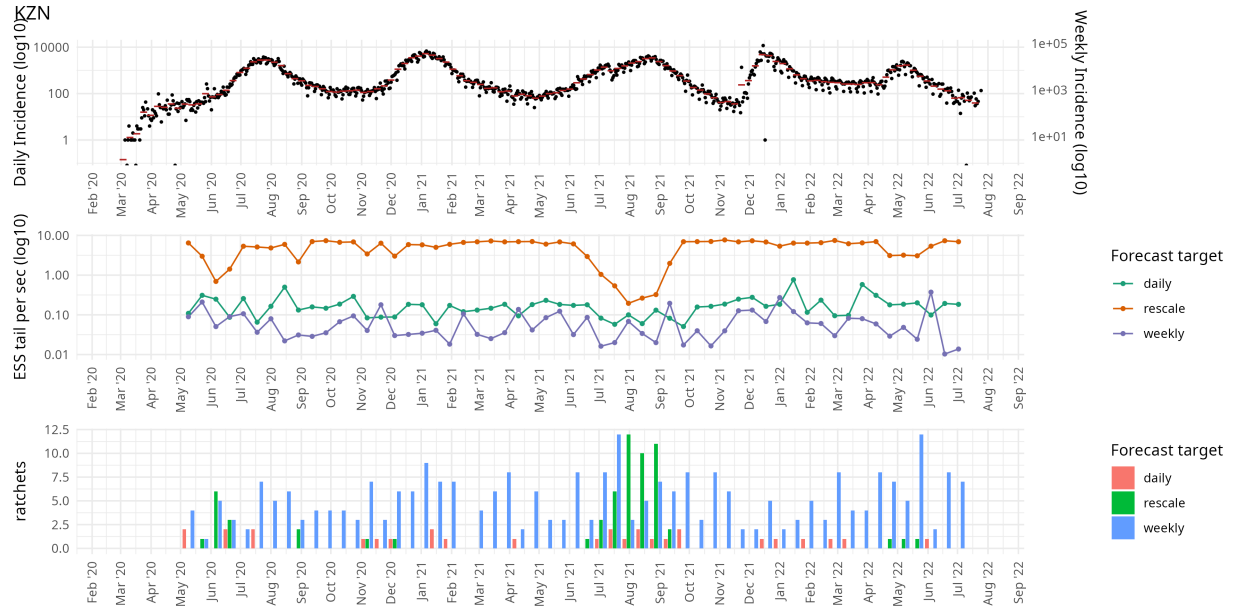


Figure 12: Computational diagnostics for the KwaZulu-Natal province (KZN). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

Limpopo (LP)

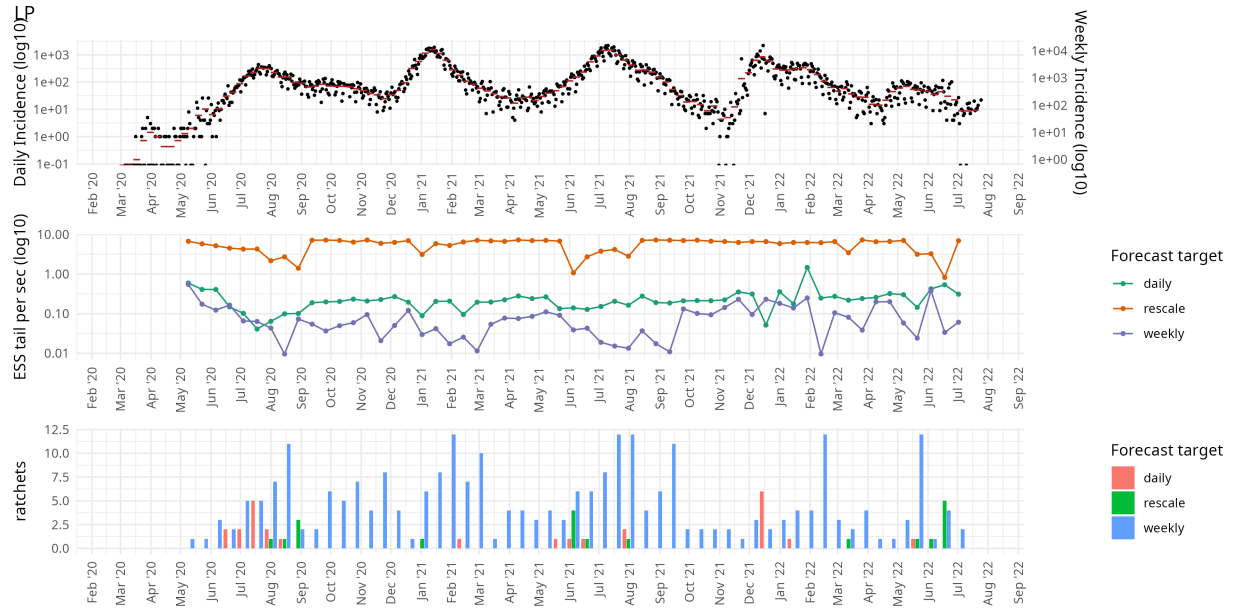


Figure 13: Computational diagnostics for the Limpopo province (LP). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

Mpumalanga (MP)

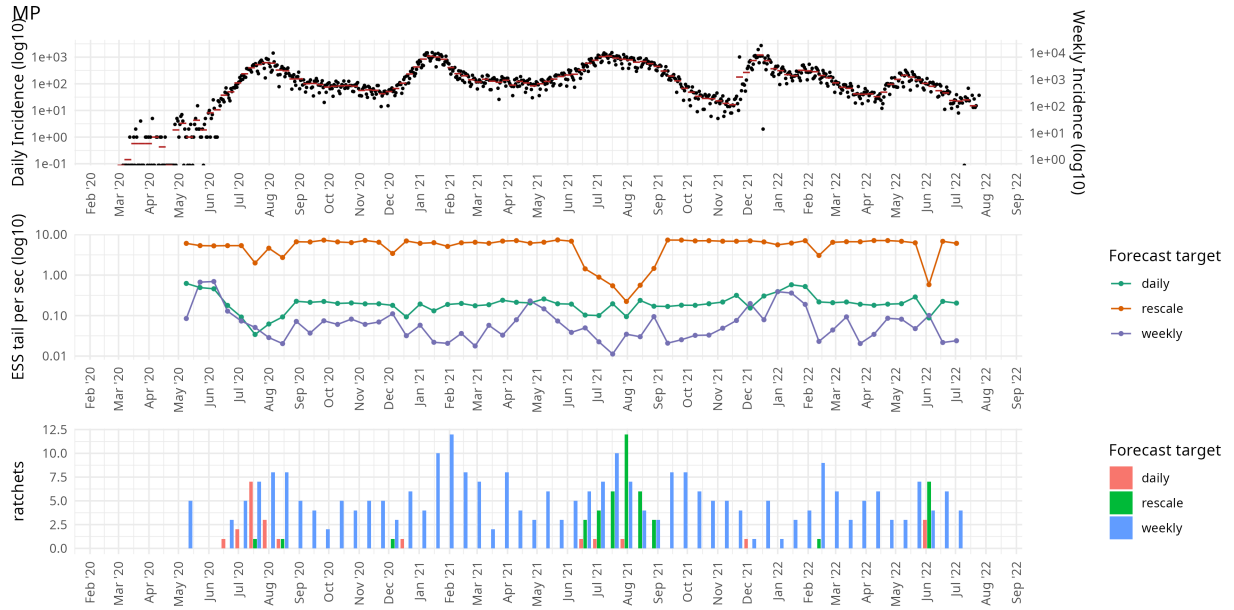


Figure 14: Computational diagnostics for the Mpumalanga province (MP). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

Northern Cape (NC)

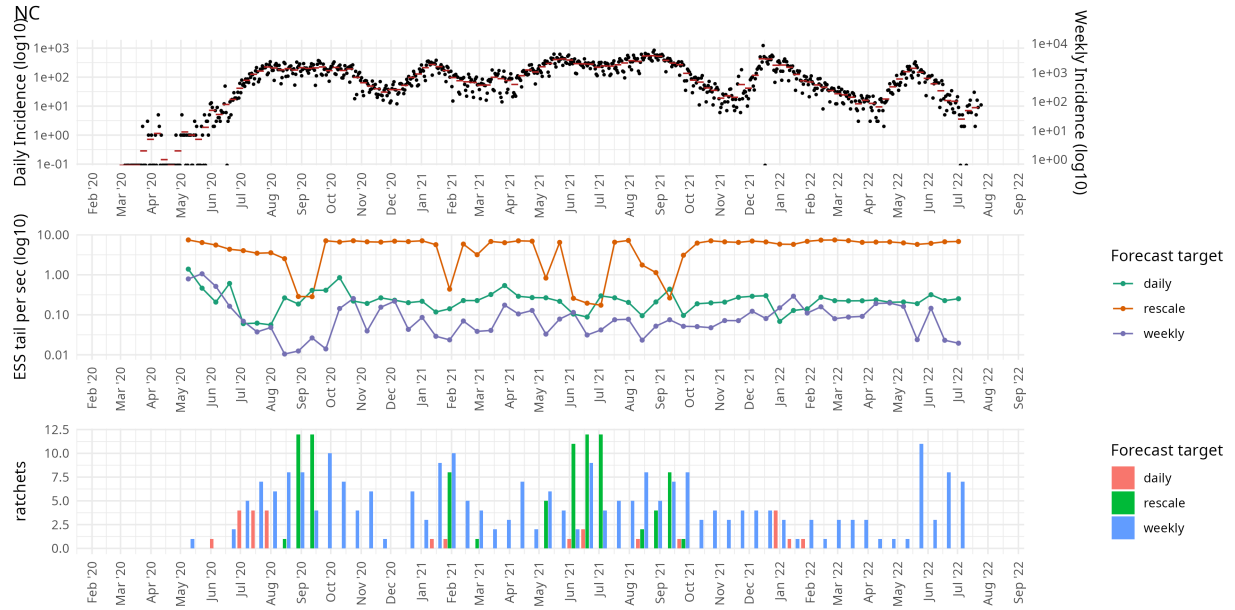


Figure 15: Computational diagnostics for the Northern Cape province (NC). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

North West (NW)

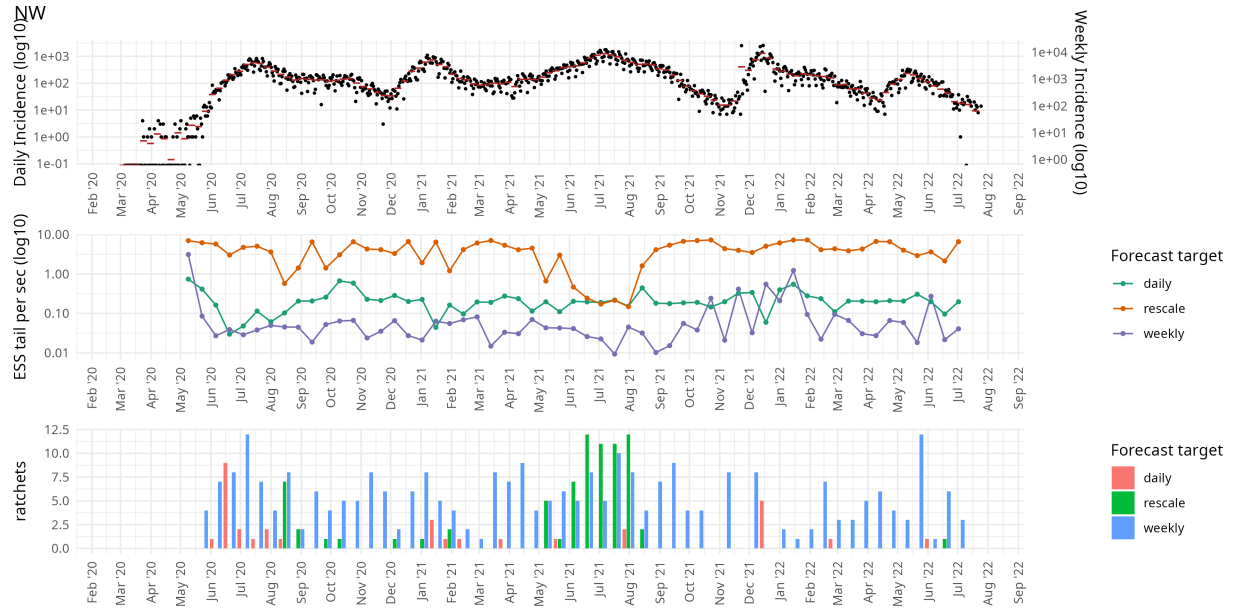


Figure 16: Computational diagnostics for the North West province (NW). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

South Africa national aggregate (RSA)

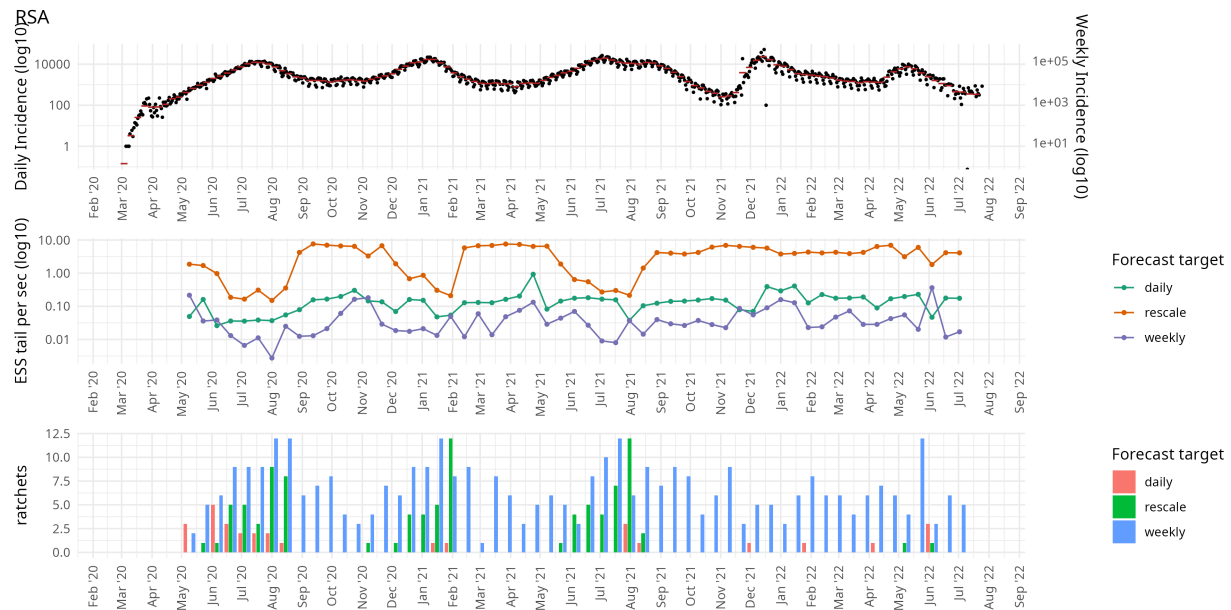


Figure 17: Computational diagnostics for the national aggregate (RSA). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.

Western Cape (WC)

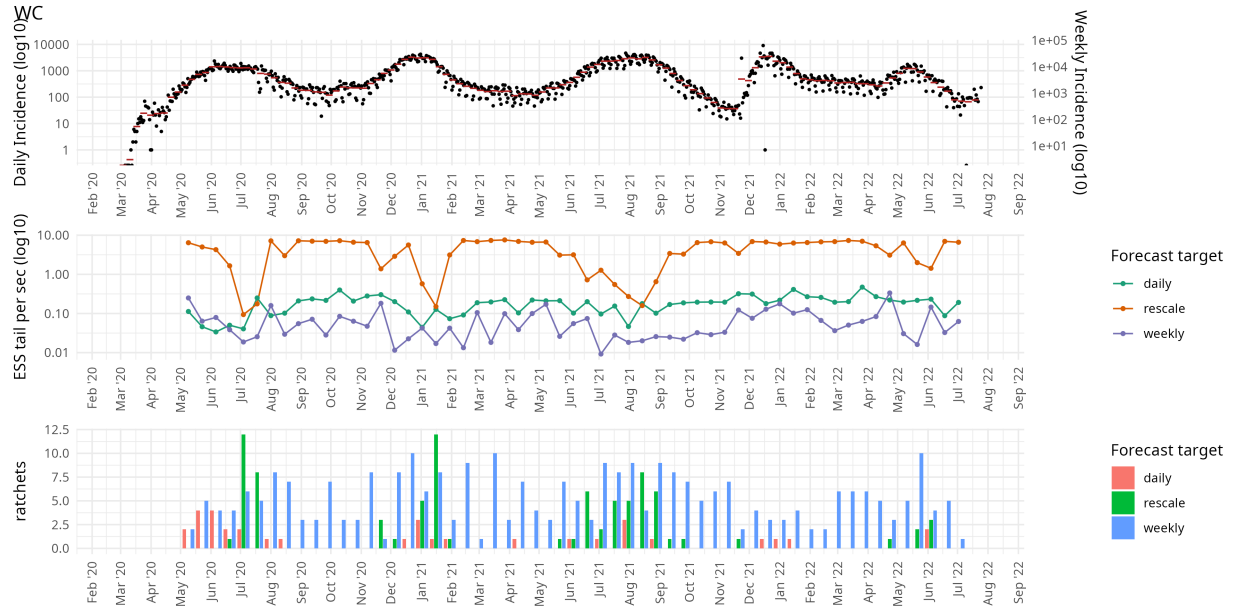


Figure 18: Computational diagnostics for the Western Cape province (WC). The top panel shows the daily and weekly COVID-19 incidence. The middle panel displays the tail Effective Sample Size (ESS) per second on a log10 scale, representing sampling efficiency for different forecast targets (colours). The bottom panel shows the number of model ratchets (adaptive refits) required to achieve convergence for each forecast slide.